Co Audit

A financial filing auditor agent that checks its own numbers and asks a human only when it is not sure.

AAPL FY2026

Revenue	478,834
Net income	checked
Operating cash flow	FLAGGED

Flag shown is from the fault demo (simulation)

MineArena 1.0 | Code Commit, BIET Davangere | PS-01 | Python

The problem and our answer

The problem

- Auditors copy numbers out of filings by hand
- It is slow, and a wrong digit is easy to miss
- An AI that guesses silently is worse than no AI

Our answer

- Read real SEC filings and pull 8 key numbers
- Check every number with accounting tests
- Score confidence; pause and ask a person when unsure
- Log every human decision in a record that cannot be edited

How it works, end to end

1

Fetch

SEC EDGAR, cached

2

Parse

HTML tables to numbers

3

Extract

8 metrics by rules

4

Second reading

LLM, grounded in the table

5

Check and score

6 checks, confidence

6

Human review

Pause, approve or fix

7

Verify

Compare with SEC XBRL

Every human decision is written to an append-only audit log, and the final numbers are checked against the SEC's official data.

Six checks and an earned confidence score

Accounting checks

- Assets = liabilities + equity
- Segments add up to the total
- Year-on-year change within limit
- Net income ties out
- Gross profit identity
- Operating income identity

Confidence score

35% how the value was read

15% unit is clear

50% checks passed

Flagged for a human if the score is below 0.85 or any check fails.

The LLM reads too, but never blindly

1. Gets the three statement tables as text
2. Names the row and the number for each metric
3. Grounding check: the number must be in that row's latest year, or it is rejected

Agrees with the rules

One more passed check

Disagrees

Flagged for a human, with the reason

No key, or the service is down

The run continues on rules and checks

Tested with a simulated model service (24 tests). Accuracy on a real model is the next measurement.

Human review and an append-only log

Pause and resume

- LangGraph interrupt() stops the run
- State saved in a SQLite checkpoint
- Reviewer approves, edits or rejects
- Resumes without repeating any step
- A step ledger proves it

Append-only audit log

- Triggers reject UPDATE and DELETE
- The app connection denies DROP and ALTER
- Each row: who, before, after, decision, time
- Simulated faults are tagged

Limit: someone editing the database file with outside tools could still remove the triggers.

Proof on real filings

58 / 58

values match the SEC's official data

8 companies

0 wrong

among 36 auto-approved values

6 companies

90%

of deliberate mistakes caught

synthetic test

202

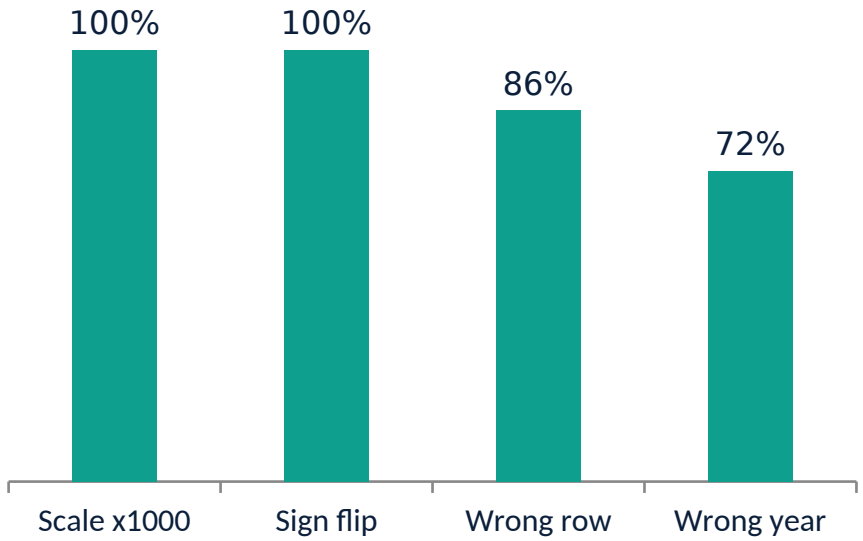
automated tests pass

all green

Apple, Microsoft, Nvidia, Walmart, Tesla, Costco, Amazon, Coca-Cola. Of 11 flagged values, 9 were real year-on-year changes and 2 were values we derived ourselves.

Catching mistakes

Injected mistakes caught (%)



Fault demo (simulation)

1. One value is made 1,000 times too big
2. Our checks catch it; the run pauses
3. A reviewer corrects it; the run resumes

Apple accuracy: 88% to 100%

Labelled simulated in the audit log

The backend API

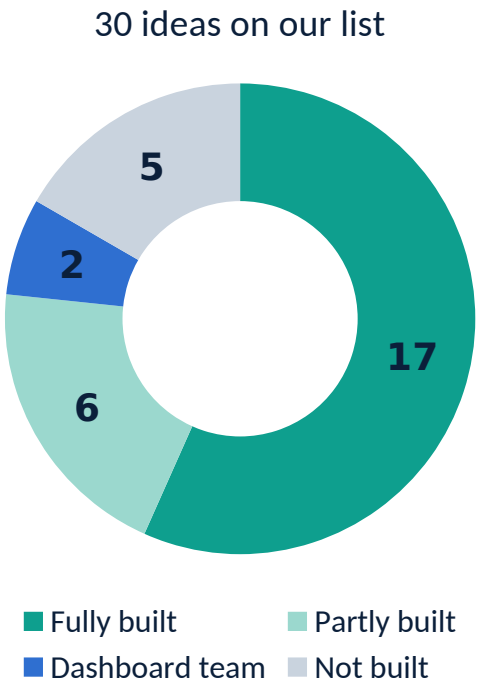
Method	Path	Purpose
POST	/runs	Start an audit run
GET	/runs/{id}	Status, metrics, checks
GET	/reviews	Pending and done reviews
POST	/reviews/{id}/decision	Approve, edit or reject
GET	/audit	Append-only audit log
GET	/report/{id}	Accuracy before and after
GET/PUT	/config	Read or change thresholds
GET	/health	Service check

Built with

- FastAPI and Pydantic
- LangGraph and SQLite
- BeautifulSoup and lxml
- Python only

The dashboard calls these endpoints.

What is built, and what is next



Built

Everything the brief requires on the backend, plus calibration, error injection, the fault switch, cached offline mode and the LLM grounding check.

Next, and honest limits

MCP tools, adaptive thresholds and two-person approval. Rules were tuned on 8 companies. Real-model LLM accuracy is still to be measured.